

How-To Guide: Analyst

Role: Analyst **Modules:** Analytics, Reliability Engineering, Reporting, Defect Elimination **Pages:** Page 5 (Analytics), Page 13 (Reliability Engineering), Page 15 (Reports & Data), Page 17 (Defect Elimination)

Role Description

Analysts are responsible for monitoring equipment health, tracking KPIs, predicting failures using Weibull analysis, identifying bad actors through Jack-Knife and Pareto analyses, reviewing planning and defect elimination KPIs, generating reports, and conducting root cause analyses. They transform data into actionable insights for maintenance optimization.

Prerequisites

Requirement	Details
API server running	<code>uvicorn api.main:app --reload</code> on <code>http://localhost:8000</code>
Streamlit app running	<code>streamlit run streamlit_app/Home.py</code> on <code>http://localhost:8501</code>
Database seeded	Full seed data including equipment hierarchy, failure modes, and work orders
Historical data	Failure intervals and work order history for Weibull and KPI calculations
Domain knowledge	Understanding of reliability engineering metrics (MTBF, MTTR, OEE, Weibull)

User Story AN-01: View Equipment Health Scores

Goal: Calculate and visualize the composite health score for a specific piece of equipment based on criticality, backlog, and strategy coverage.

Step-by-Step Instructions

1. **Navigate to Page 5 -- Analytics** - Open the Streamlit sidebar and click **Analytics**. - The page has three tabs: **Asset Health**, **KPIs**, and **Weibull**.

2. **Select Equipment (Asset Health Tab)** - The **Asset Health** tab is selected by default. - Use the **Select Equipment** dropdown to choose an equipment item.
3. **Configure Health Parameters (left column) - Risk Class:** Select the equipment's criticality class:
 - **I_LOW** , **II_MEDIUM** , **III_HIGH** , **IV_CRITICAL**
 - **Pending Backlog Hours:** Enter the outstanding maintenance backlog (default: 50.0).
 - **Total Failure Modes:** Number of identified failure modes for this equipment (default: 10).
 - **FM with Strategy:** Number of failure modes that have an assigned maintenance strategy (default: 8).
4. **Calculate Health Score** - Click **Calculate Health**. - The results display in the right column:
 - A gauge chart showing the **Composite Score** (0-100).
 - The **Health Class** label (GOOD, FAIR, POOR, CRITICAL).
 - A list of **Recommendations** for improvement.

API Endpoint Called

```
POST /api/v1/analytics/health-score
{
  "node_id": "uuid-...",
  "plant_id": "OCP-JFC1",
  "equipment_tag": "BRY-SAG-ML-001",
  "risk_class": "III_HIGH",
  "pending_backlog_hours": 50.0,
  "total_failure_modes": 10,
  "fm_with_strategy": 8
}
```

Expected Output

```
{
  "composite_score": 72.5,
  "health_class": "FAIR",
  "component_scores": {
    "criticality": 75,
    "backlog": 60,
    "strategy_coverage": 80,
    "alert_status": 85
  },
  "recommendations": [
    "Address pending backlog to improve score",
    "Define strategies for remaining 2 failure modes"
  ]
}
```

Error Handling

Issue	Solution
"Cannot connect to API"	Start the API server.
No equipment in dropdown	Seed the database first (Page 7).
Score seems inaccurate	Verify input parameters match actual equipment data.

User Story AN-02: Monitor KPIs

Goal: Calculate and visualize key maintenance KPIs including MTBF, MTTR, availability, and OEE.

Step-by-Step Instructions

- Navigate to Page 5 -- Analytics, KPIs Tab** - Switch to the **KPIs** tab.
- Enter KPI Parameters (left column)** - **Plant ID:** Enter the plant identifier (default: `OCP-JFC1`). - **Total Period Hours:** Total hours in the analysis period (default: 8760 for one year). - **Total Downtime Hours:** Accumulated downtime hours (default: 120.0).
- Calculate KPIs** - Click **Calculate KPIs**. - Results display in the right column:
 - A bar chart showing all calculated KPIs.
 - Individual metric cards for each KPI value.

API Endpoint Called

```
POST /api/v1/analytics/kpis
{
  "plant_id": "OCP-JFC1",
  "total_period_hours": 8760.0,
  "total_downtime_hours": 120.0
}
```

Expected Output

KPI metrics including: - **Availability:** (Total - Downtime) / Total (e.g., 98.6%) - **MTBF:** Mean Time Between Failures - **MTTR:** Mean Time To Repair - **OEE:** Overall Equipment Effectiveness

Error Handling

Issue	Solution
All KPIs show 0	Ensure downtime and period hours are set to non-zero values.
Missing KPI values	Some KPIs require failure date data. Provide failure history.

User Story AN-03: Analyze Weibull Failure Prediction

Goal: Fit a Weibull distribution to historical failure intervals to predict future failures and determine the optimal replacement interval.

Step-by-Step Instructions

1. **Navigate to Page 5 -- Analytics, Weibull Tab** - Switch to the **Weibull** tab.
2. **Enter Failure Data - Failure Intervals:** Enter comma-separated values representing days between failures (e.g., **120, 180, 95, 210, 150**). Minimum 3 intervals required. - **Current Age:** Enter the current operating age in days since last failure (default: 100.0).
3. **Fit Weibull Distribution** - Click **Fit Weibull**. - Results display:
 - **Beta (Shape):** Indicates failure pattern ($\beta < 1$ = early life, $\beta = 1$ = random, $\beta > 1$ = wear-out).
 - **Eta (Scale):** Characteristic life in the same time unit as inputs.
 - **R-squared:** Goodness of fit (closer to 1.0 is better).
 - A reliability curve $R(t)$ vs Time is plotted showing the probability of survival over time.

API Endpoint Called

```
POST /api/v1/analytics/weibull-fit
{
  "failure_intervals": [120, 180, 95, 210, 150]
}
```

Expected Output

```
{
  "beta": 2.45,
  "eta": 165.3,
  "gamma": 0,
  "r_squared": 0.9823
}
```

Interpretation: - Beta > 1 indicates wear-out failures -- time-based replacement is appropriate. - The characteristic life (eta) suggests failures cluster around 165 days.

Error Handling

Issue	Solution
"Need at least 3 intervals"	Provide 3 or more comma-separated failure intervals.
R-squared is low (< 0.8)	Data may not follow a Weibull distribution well. Check for outliers or data quality issues.
Beta is very close to 1	Failures are essentially random. CBM may be more appropriate than TBM.

User Story AN-04: Jack-Knife Analysis

Goal: Identify bad-actor equipment using Jack-Knife quadrant analysis that classifies equipment by failure frequency and downtime impact.

Step-by-Step Instructions

- Navigate to Page 13 -- Reliability Engineering** - Open the Streamlit sidebar and click **Reliability Engineering**. - Click the **Bad Actors** tab, then the **Jack-Knife** sub-tab.
- Configure Analysis - Plant ID:** Enter the plant identifier (default: `OCP-JFC1`).
- Run Jack-Knife Analysis** - Click **Run Jack-Knife** (blue primary button). - The system uses sample equipment data to demonstrate the analysis.
- Review Results** - Four metric cards show equipment counts by quadrant:
 - Acute:** High frequency, high downtime -- top priority
 - Chronic:** High frequency, low downtime -- address root causes
 - Complex:** Low frequency, high downtime -- investigate when they fail
 - Controlled:** Low frequency, low downtime -- acceptable performance
 - A scatter plot (Jack-Knife chart) visualizes each equipment point in the four quadrants.

API Endpoint Called

```
POST /api/v1/reliability/jackknife/analyze
{
  "plant_id": "OCP-JFC1",
  "equipment_data": [
    {
      "equipment_id": "EQ-1",
      "equipment_tag": "BRY-SAG-001",
      "failure_count": 15,
      "total_downtime_hours": 120,
      "operating_hours": 8760
    }
  ]
}
```

Expected Output

```
{
  "acute_count": 1,
  "chronic_count": 1,
  "complex_count": 1,
  "controlled_count": 2,
  "points": [
    {
      "equipment_id": "EQ-1",
      "equipment_tag": "BRY-SAG-001",
      "failure_rate": 1.71,
      "mean_downtime": 8.0,
      "quadrant": "ACUTE"
    }
  ]
}
```

Error Handling

Issue	Solution
Analysis returns error	Check that equipment data array is not empty. Verify operating hours > 0.
All equipment in one quadrant	Adjust thresholds or provide data with more variation.

User Story AN-05: Pareto Analysis

Goal: Perform Pareto (80/20) analysis to identify the top contributors to failures, cost, or downtime.

Step-by-Step Instructions

1. **Navigate to Page 13 -- Reliability Engineering** - Click the **Bad Actors** tab, then the **Pareto** sub-tab.
2. **Configure Analysis - Plant ID:** Enter the plant identifier. - **Metric Type:** Select the analysis dimension:
 - **failures** -- Rank by failure count
 - **cost** -- Rank by failure cost
 - **downtime** -- Rank by downtime hours
3. **Run Pareto Analysis** - Click **Run Pareto** (blue primary button).
4. **Review Results** - A metric shows: "X / Y bad actors" (the 20% causing 80% of the problem). - A Pareto chart with bars (individual contribution) and a cumulative line. - Items above the 80% threshold are highlighted as bad actors.

API Endpoint Called

```
POST /api/v1/reliability/pareto/analyze
{
  "plant_id": "OCP-JFC1",
  "metric_type": "cost",
  "records": [
    { "equipment_id": "EQ-1", "equipment_tag": "BRY-001", "cost": 50000, "downtime_hours": 120 }
  ]
}
```

Expected Output

```
{
  "bad_actor_count": 3,
  "items": [
    {
      "equipment_id": "EQ-1",
      "equipment_tag": "BRY-001",
      "value": 50000,
      "cumulative_pct": 38.5,
      "is_bad_actor": true
    }
  ]
}
```

Error Handling

Issue	Solution
Empty results	Ensure records array has data. At least 2 records are needed for meaningful analysis.
No bad actors identified	All equipment has similar contribution. No clear 80/20 split exists.

User Story AN-06: Review Planning KPIs

Goal: Calculate and review the 11 planning KPIs defined by the GFSN REF-14 standard.

Step-by-Step Instructions

1. **Navigate to Page 17 -- Defect Elimination** - Open the Streamlit sidebar and click **Defect Elimination**.
- Switch to the **Planning KPIs** tab.
2. **Enter KPI Input Data** - Fill in all planning metrics across the input columns:
 - **WOs Planned / Completed:** Work order execution data
 - **Man-hours Planned / Actual:** Labor tracking
 - **PM Planned / Executed:** Preventive maintenance compliance
 - **Backlog Hours / Weekly Capacity:** Backlog health
 - **Corrective / Proactive WOs:** Maintenance mix
 - **Schedule Compliance:** Planned vs. executed
 - **Release Horizon:** Planning lead time in days
 - **Pending / Total Notices:** Notice processing efficiency
 - **Scheduled / Total Capacity:** Capacity utilization
3. **Calculate Planning KPIs** - Click **Calculate Planning KPIs** (blue primary button).
4. **Review Results** - Overall health status: "Overall Health: GOOD -- X/11 on target" - A data table showing each KPI with:
 - Status indicator (on target / off target)
 - KPI name, value, target, and unit
 - A radar chart showing all 11 KPIs vs. their targets.

API Endpoint Called

```
POST /api/v1/rca/planning-kpis/calculate
{
  "plant_id": "OCP-JFC1",
  "period_start": "2025-03-01",
  "period_end": "2025-03-08",
  "wo_planned": 100,
  "wo_completed": 92,
  "manhours_planned": 800.0,
  "manhours_actual": 780.0,
  "pm_planned": 50,
  "pm_executed": 48,
  "backlog_hours": 300.0,
  "weekly_capacity_hours": 200.0,
  ...
}
```


Expected Output

```
{
  "overall_health": "GOOD",
  "on_target_count": 9,
  "kpis": [
    {
      "name": "wo_completion_rate",
      "value": 92.0,
      "target": 90.0,
      "unit": "%",
      "status": "ON_TARGET"
    }
  ]
}
```

Error Handling

Issue	Solution
KPI values all null	Ensure numeric inputs are non-zero. Check that planned values > 0.
Low on-target count	Review individual KPIs below target and investigate root causes.

User Story AN-07: Review DE KPIs

Goal: Calculate and review the 5 Defect Elimination program KPIs per GFSN REF-15.

Step-by-Step Instructions

1. **Navigate to Page 17 -- Defect Elimination** - Switch to the **DE KPIs** tab.
2. **Enter DE KPI Input Data** - **Events Reported / Required:** DE event capture rate - **Meetings Held / Required:** Team engagement tracking - **Actions Implemented / Planned:** Action completion rate - **Savings Achieved / Target:** Financial impact tracking - **Failures Current / Previous:** Failure rate trend
3. **Calculate DE KPIs** - Click **Calculate DE KPIs** (blue primary button).
4. **Review Results** - **Program Maturity Level:** INITIATING, DEVELOPING, ESTABLISHED, or OPTIMIZING - **Program Score:** 0-100 gauge chart - Individual KPI table with status indicators - **Strengths:** Areas performing well - **Improvement Areas:** Areas needing attention

API Endpoint Called

```
POST /api/v1/rca/de-kpis/calculate
{
  "plant_id": "OCP-JFC1",
  "period_start": "2025-02-01",
  "period_end": "2025-03-01",
  "events_reported": 18,
  "events_required": 20,
  "meetings_held": 9,
  "meetings_required": 10,
  "actions_implemented": 14,
  "actions_planned": 16,
  "savings_achieved": 85000.0,
  "savings_target": 100000.0,
  "failures_current": 8,
  "failures_previous": 12
}
```

Expected Output

```
{
  "kpis": {
    "kpis": [
      { "name": "event_reporting_rate", "value": 90.0, "target": 85, "status": "ON_TARGET" }
    ]
  },
  "health": {
    "maturity_level": "ESTABLISHED",
    "program_score": 78,
    "strengths": ["Event reporting rate exceeds target"],
    "improvement_areas": ["Increase savings realization"]
  }
}
```

Error Handling

Issue	Solution
Division by zero errors	Ensure "required" and "target" fields are greater than zero.
Low maturity score	Review improvement areas and create action plans for each gap.

User Story AN-08: Generate Reports

Goal: Generate weekly, monthly, or quarterly maintenance reports and manage report history.

Step-by-Step Instructions

1. **Navigate to Page 15 -- Reports & Data** - Open the Streamlit sidebar and click **Reports & Data**. - The page has four tabs: **Generate**, **History**, **Import**, and **Export**.
2. **Generate a Report (Generate Tab)** - Enter the **Plant ID** (default: `OCP-JFC`). - Select **Report Type**: Weekly, Monthly, or Quarterly. - For **Weekly Reports**:
 - Set Week Number and Year
 - Enter Safety Incidents count and Backlog Hours
 - Click **Generate Weekly Report**
 - For **Monthly Reports**:
 - Set Month and Year
 - Click **Generate Monthly Report**
 - For **Quarterly Reports**:
 - Set Quarter and Year
 - Click **Generate Quarterly Report**
3. **View Report History (History Tab)** - Switch to the **History** tab. - Filter by report type: `All`, `WEEKLY_MAINTENANCE`, `MONTHLY_KPI`, `QUARTERLY_REVIEW`. - Expand any report to see details. - Click **View** to see the full report JSON.
4. **Export Data (Export Tab)** - Switch to the **Export** tab. - Select export type: `equipment`, `kpis`, `schedule`, `report`. - Click **Prepare Export**. - Review the generated sheets with row counts and headers.

API Endpoints Called

```
POST /api/v1/reporting/reports/weekly
  { "plant_id": "OCP-JFC", "week": 12, "year": 2025, "safety_incidents": 0, "backlog_hours": 150 }
POST /api/v1/reporting/reports/monthly
  { "plant_id": "OCP-JFC", "month": 3, "year": 2025 }
POST /api/v1/reporting/reports/quarterly
  { "plant_id": "OCP-JFC", "quarter": 1, "year": 2025 }
GET /api/v1/reporting/reports?plant_id=OCP-JFC&report_type=MONTHLY_KPI
GET /api/v1/reporting/reports/{report_id}
POST /api/v1/reporting/export
  { "export_type": "kpis" }
```

Expected Output

Weekly report response:

```
{
  "report_id": "uuid-...",
  "report_type": "WEEKLY_MAINTENANCE",
  "plant_id": "OCP-JFC",
  "period_start": "2025-03-17",
  "period_end": "2025-03-23",
  "generated_at": "2025-03-23T14:00:00",
  "content": { "safety_incidents": 0, "backlog_hours": 150, ... }
}
```

Error Handling

Issue	Solution
Report generation fails	Check API connection. Ensure plant ID is valid.
No reports in history	Generate at least one report first.
Export shows empty sheets	Ensure the database contains data for the selected export type.

User Story AN-09: Conduct Root Cause Analysis (RCA)

Goal: Create and manage root cause analyses for failure events using the structured RCA framework with 5W+2H methodology.

Step-by-Step Instructions

- Navigate to Page 17 -- Defect Elimination** - The **RCA** tab is selected by default.
- Create a New RCA (left column)** - Enter:
 - **Plant ID:** Plant where the event occurred (default: **OCP-JFC1**)
 - **Event Description:** Detailed description of the failure event
 - **Equipment ID** (optional): Related equipment identifier
 - **Max Consequence** (1-5 slider): Severity of the consequence
 - **Frequency** (1-5 slider): How often this type of event occurs
 - **Team Members:** Comma-separated list of investigators
 - Click **Create RCA** (blue primary button).
 - The system determines the RCA level based on consequence x frequency:
 - **Level 1:** Simple 5W+2H analysis
 - **Level 2:** Cause-and-Effect (Fishbone) analysis
 - **Level 3:** Full RCA with Fault Tree Analysis

3. **Review Active RCAs (right column)** - Summary metrics: Total, Open, Under Investigation, Completed. - A data table lists all RCA analyses. - A chart shows the distribution of RCA levels.
4. **Run 5W+2H Analysis (5W+2H Tab)** - Switch to the **5W+2H** tab. - Enter the **RCA Analysis ID** from the created RCA. - Fill in the seven structured fields:
 - **WHAT**: Problem description and goal
 - **WHEN**: Timing and schedule
 - **WHERE**: Location and implementation area
 - **WHO**: Skills required and responsible person
 - **WHY**: Traceability and justification
 - **HOW**: Manifestation and approach
 - **HOW MUCH**: Quantification and cost impact
 - Click **Submit 5W+2H Analysis**.
 - A formatted report is generated and displayed.

API Endpoints Called

```
POST /api/v1/rca/analyses
{
  "event_description": "SAG mill bearing failure...",
  "plant_id": "OCP-JFC1",
  "equipment_id": "EQ-SAG-001",
  "max_consequence": 4,
  "frequency": 3,
  "team_members": ["J. Garcia", "M. Benali"]
}
GET /api/v1/rca/analyses/summary?plant_id=OCP-JFC1
GET /api/v1/rca/analyses?plant_id=OCP-JFC1
POST /api/v1/rca/analyses/{analysis_id}/5w2h
{ "what": "...", "when": "...", "where": "...", "who": "...", "why": "...", "how": "...",
  "how_much": "..." }
```

Expected Output

RCA creation:

```
{
  "analysis_id": "uuid-...",
  "level": "LEVEL_2",
  "status": "OPEN",
  "team_requirements": {
    "description": "Cause-and-Effect (Fishbone) analysis required",
    "minimum_team_size": 3
  }
}
```

Error Handling

Issue	Solution
"Describe the event" warning	The event description field is required.
"API not available"	Start the API server with <code>uvicorn api.main:app --reload</code> .
5W+2H fails with 404	Ensure the analysis ID is correct and the RCA exists.

Quick Reference: Analyst Workflow

Page 5: Analytics

```
+-----+
| AN-01: Health Scores |
| AN-02: KPIs (MTBF/OEE) |
| AN-03: Weibull Predict |
+-----+
```

Page 15: Reports & Data

```
+-----+
| AN-08: Generate Reports |
| View history & export |
+-----+
```

Page 13: Reliability

```
+-----+
| AN-04: Jack-Knife |
| AN-05: Pareto Analysis |
+-----+
```

Page 17: Defect Elimination

```
+-----+
| AN-06: Planning KPIs |
| AN-07: DE KPIs |
| AN-09: Conduct RCA |
+-----+
```